1. Problem Statement, Objectives, and Assumptions

Problem Statement

Educational institutions often enforce a mandatory attendance policy (typically 75%). Students falling short of this threshold face academic penalties or exam restrictions. Manually calculating how many consecutive classes need to be attended to recover the required percentage can be complex and prone to errors. A simple tool is needed to compute the exact number of consecutive classes required to reach the target attendance.

Objectives

- Accept user inputs for the number of classes attended and the total classes held.
- Validate input parameters to ensure logical consistency (e.g., attended classes cannot exceed total classes, and total classes must be positive).
- Calculate and display the current attendance percentage.
- Compute the exact number of future consecutive classes a student must attend to hit the 75% threshold.

Assumptions

- The target attendance percentage is fixed at 75%.

- Every future class counted in the recovery plan is successfully attended (both attended count and total classes increment by 1).
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- Inputs are valid integers representing counts.
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2. Algorithm, Flowchart, and Pseudocode

Algorithm

1. Read the number of classes attended and total classes held.
2.
3. Check if total <= 0, attended < 0, or attended > total. If true, display "Invalid attendance details" and terminate.
4.
5. Calculate the current attendance percentage as (attended / total) * 100.
6.
7. Initialize a counter extra = 0.
8.
9. Loop while (attended + extra) / (total + extra) * 100 < 75:
10.
    - Increment extra by 1.
    -
11. Print the current attendance percentage, the number of extra classes needed, and the target percentage.
12.

Flowchart Representation

Pseudocode

3. Python Source Code and Test Cases

Python Source Code

```
            int input "Classes attended: "
        int input "Total classes held: "
          75
if          0 or              0 or
        "Invalid attendance details." else
                                  100
          0

  while                                 100
              1

        "Current attendance:"  round            2   "%"
        "More classes to attend:"
        "Target attendance:"          "%"
```

(Source code structure adapted from).

Test Cases & Sample Output

Test Case ID	Scenario Description	User Inputs	Expected Output / Behavior
TC-01	Attendance below 75%	Attended: 30	

Total: 50 | Current attendance: 60.0%

More classes to attend: 10. | | **TC-02** | Attendance already above 75% | Attended: 40

Total: 50 | Current attendance: 80.0%

More classes to attend: 0. | | **TC-03** | Invalid Details (Attended > Total) | Attended: 55

Total: 50 | "Invalid attendance details.". |

Simulated Terminal Output (for TC-01)

4. Conclusion

The attendance recovery script effectively automates the calculation of academic attendance requirements. By combining basic arithmetic, input validation, and an iterative loop, it provides students with a precise roadmap to meet institutional criteria without manual guesswork.